

Goal :
Track overall accident trends over time and severity distribution

Safety Trends Dashboard

Target audience:
Aviation safety regulators / executive leadership (NTSB-level)

Safety Trends

Aircraft & Engine Analysis

Flight Purpose & Injury Impact

Weather Condition

All

Aircraft Damage

All

Engine Type

All

150.04K

Total Accidents

66K

Total Fatal Injuries

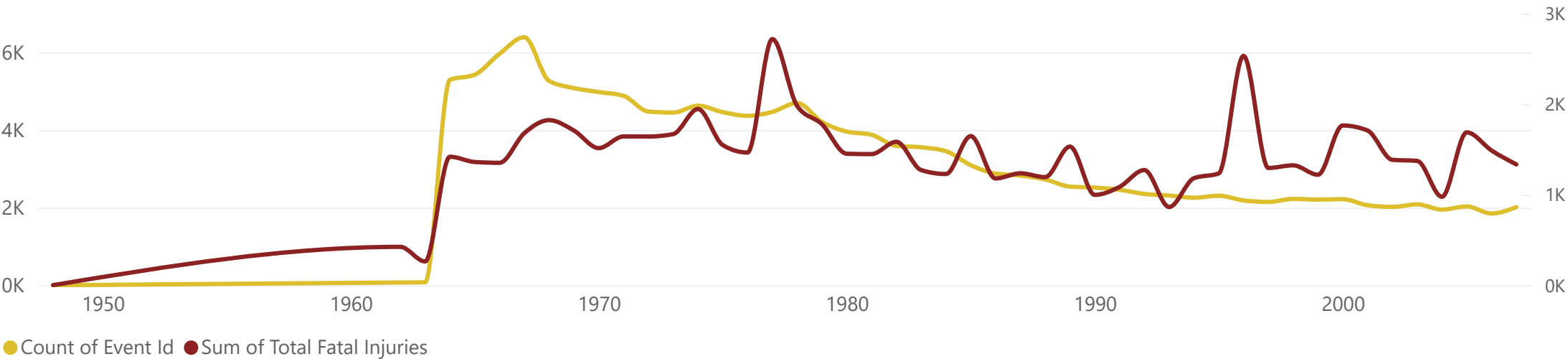
0.44

Average Fatal Injuries

0.68

Average Injuries

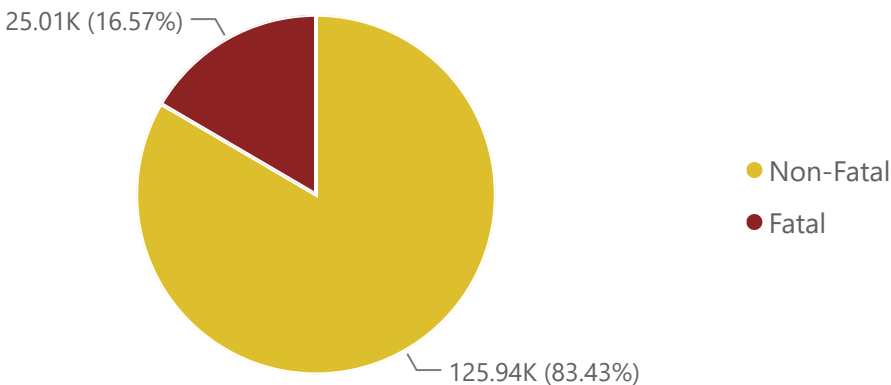
Yearly Trend of Total Accidents vs. Fatalities



Survivability Trend: Uninjured Outcomes Over Time



Accidents Distribution by Injury Severity (Fatal vs. Non-Fatal)
(Fatal vs. Non-Fatal)



Accidents with Significant Reporting Delay

(> 365 Days)

Report Publication Date

Event Date

Event Id	Year	Month	Day	Year	Month	Day
20060606X00685	2020	February	27	2006	May	31
20061220X01815	2020	February	27	2006	December	10
20070720X00969	2020	February	27	2007	July	16
20071213X01937	2020	February	27	2007	December	3
20001205X00332	2019	October	8	1999	March	14
20001208X06203	2019	June	6	1996	July	6
20001208X08803	2019	September	27	1997	September	14
20001212X24393	2019	September	23	1990	October	6
20001213X30060	2019	July	10	1987	January	15
20001214X38563	2019	June	5	1984	January	26
20001214X41363	2019	June	5	1984	October	26
20011025X02148	2019	June	17	2001	October	19
20070524X00626	2019	June	17	2007	April	12
20070711X00015	2019	June	17	2007	July	0

Goal : Understand which aircraft types/configurations are most associated with accidents.

Aircraft & Engine Analysis

Target audience:
Aircraft manufacturers & maintenance/engineering teams

Safety Trends

Aircraft & Engine Analysis

Flight Purpose & Injury Impact

Aircraft Damage

All

Amateur Built

All

Engine Type

All

4.94%

Amateur Built Rate %

10.44%

Destroyed Aircraft %

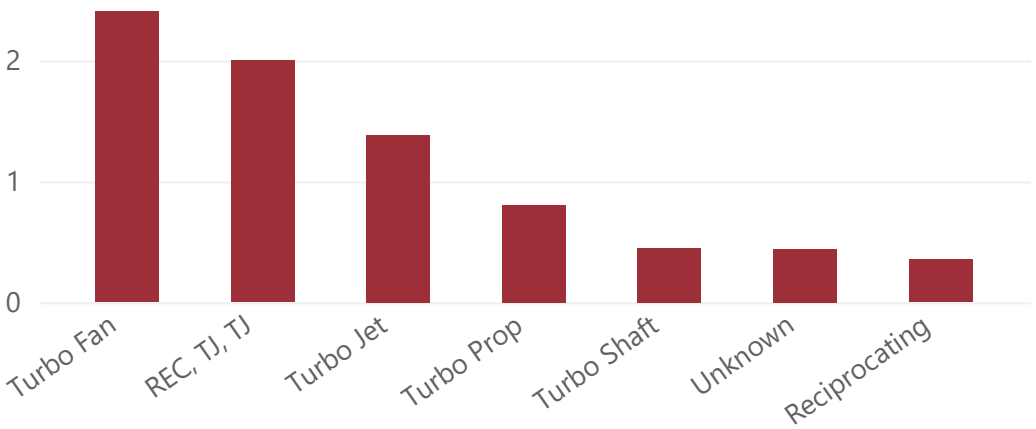
REC, TJ, TJ

First Engine Type

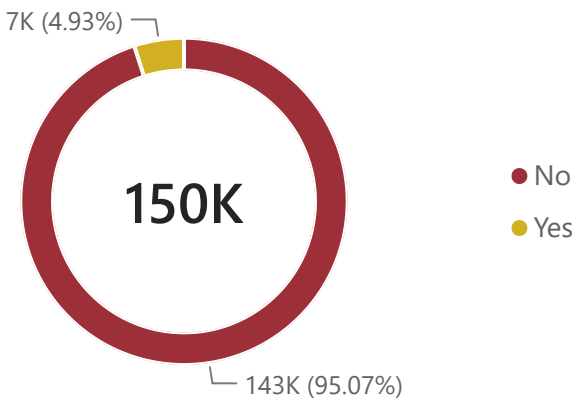
OBERLERCHNE

First Make

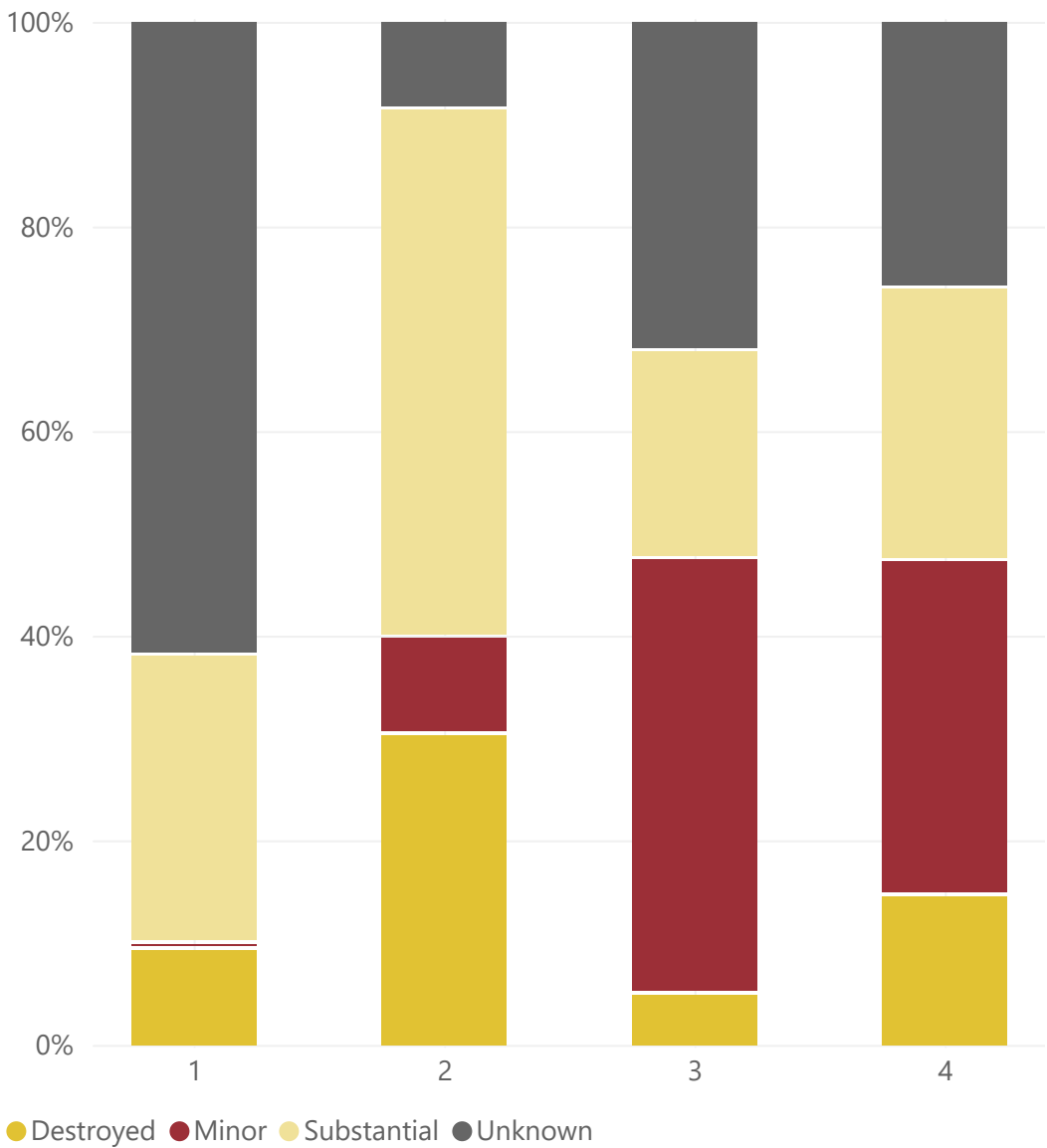
Average Fatality Rate per Accident by Engine Type



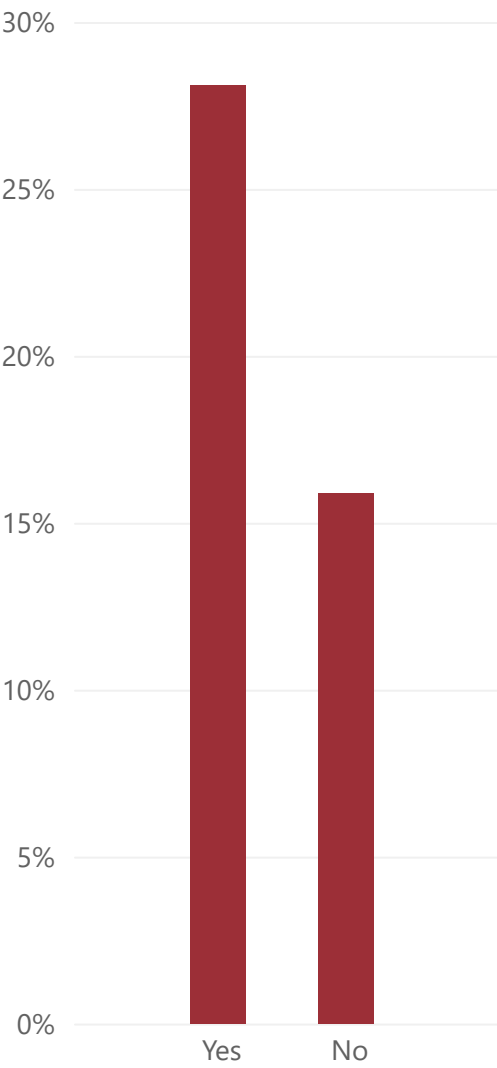
Accidents Ratio: Amateur-Built vs. Certified Aircraft



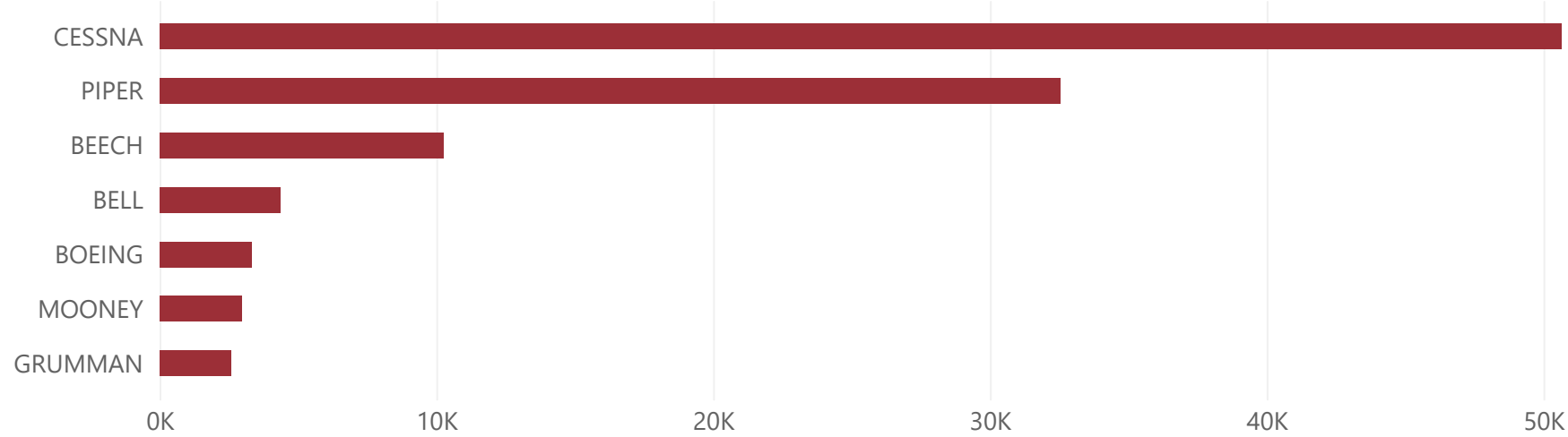
Aircraft Damage Severity Distribution by Number of Engines



Fatal Accident Rate Comparison: Amateur-Built vs. Standard



Top 5 Aircraft Models in Fatal Accident



Goal :Analyze how the purpose of the flight relates to injury outcomes, for risk/insurance assessment.

Flight Purpose & Injury Impact

Target audience:
Insurance & risk management teams

Safety Trends

Aircraft & Engine Analysis

Flight Purpose & Injury Impact

FAR Description

All

Purpose of Flight

All

Injury Severity

All

Aerial Application

First Purpose of Flight

103K

Total Injuries Count

1.95

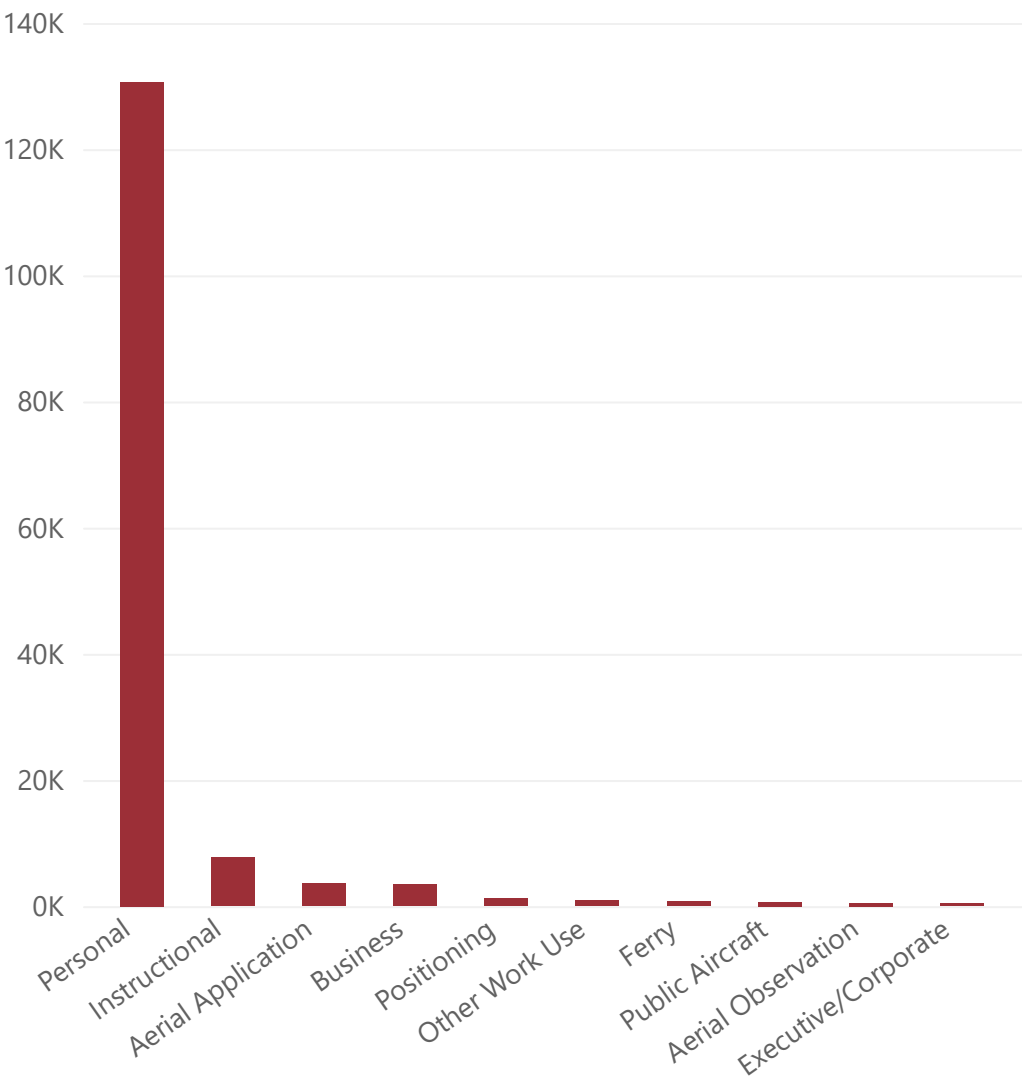
Avg Uninjured per Accident

Armed Forces

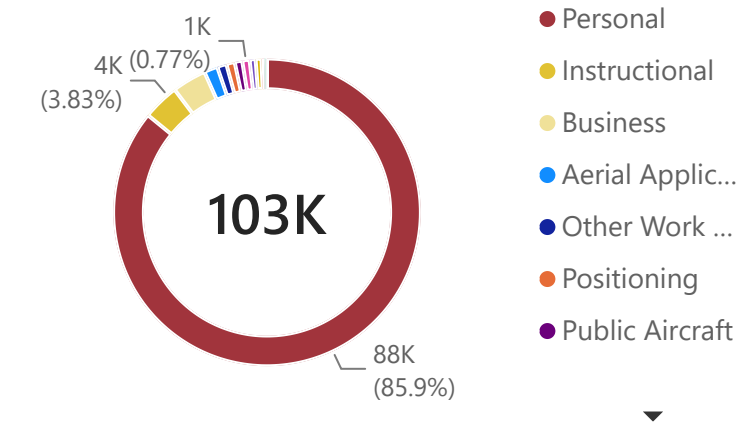
First FAR Description

Total Accidents by Purpose of Flight

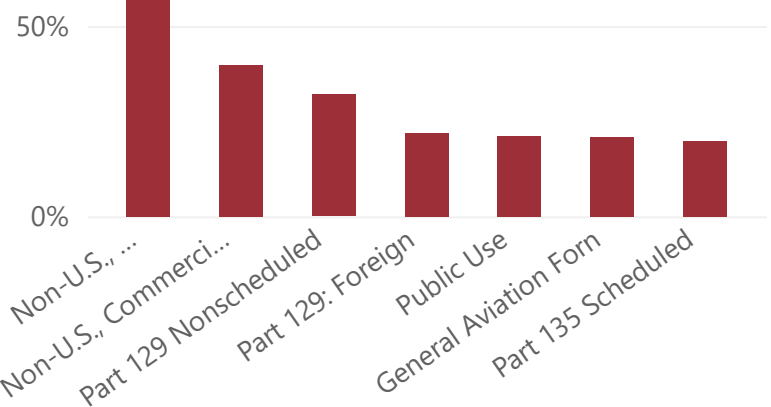
Top 10



Accidents Ratio: Amateur-Built vs. Certified Aircraft



Top 5 Aircraft Models in Fatal Accident



Aircraft Damage Severity Distribution by Number of Engines

